

Architecture Driven AI Engineering

Agentic AI Engineering Rigor, MBSE, & Continuous Delivery
Speed Through Full Lifecycle Engineering

JD Black

Sr Director, Chief Engineer
Chief Information & Digital Office (CIDO)

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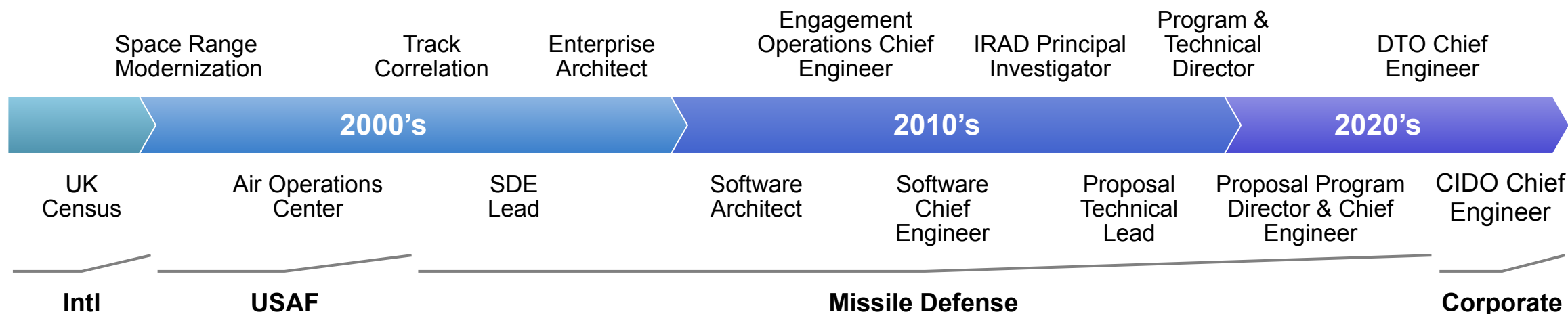
Who Am I?



JD Black

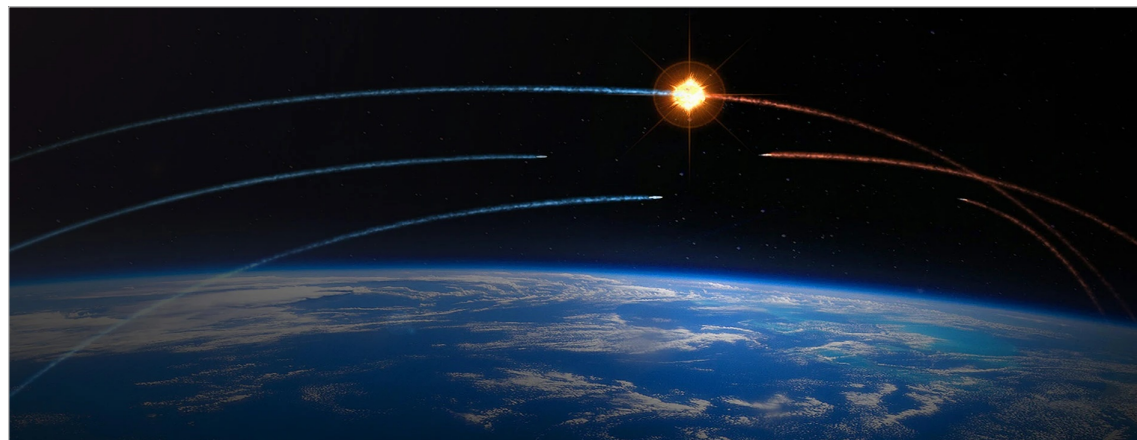
Sr Director, Chief Engineer
Northrop Grumman
Chief Information & Digital Office
www.linkedin.com/in/jondavid-black

- 2 NG Presidents Awards
- DevOps Enterprise Summit (DOES) Speaker
- ETLs Journal Author (DoD, GenAI in DevOps, Tech Leaders)
- DevOps Days Chattanooga Speaker
- Led Huntsville's DevOps for Defense Meetup for 7 Years
- First "Official" Approved NG Open-Source Project
- Quoted in NY Times
- Satellite Tool Kit (STK) User's Conference Speaker
- Built Space Range Collision Avoidance Tool
- Designed, Built, Deployed, & Operated World's Largest SAN
- Present when Jedi Knight became a "religion" in UK



Northrop Grumman Overview

<https://www.northropgrumman.com/who-we-are/the-facts>



Northrop Grumman Jobs
96K

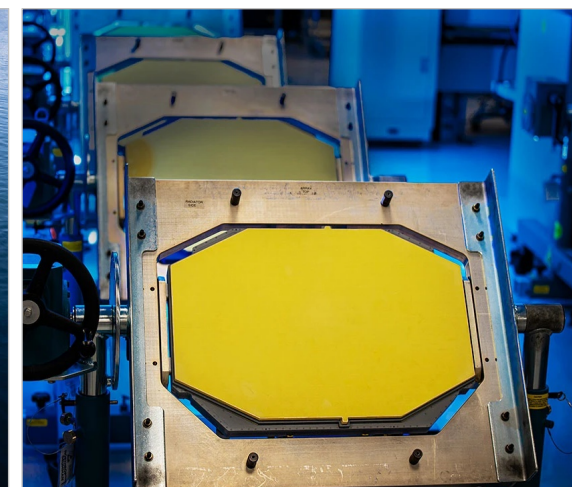
Supplier Partners
10K

Mission-Driven Engineers
43K+

US Manufacturing
30M sq ft

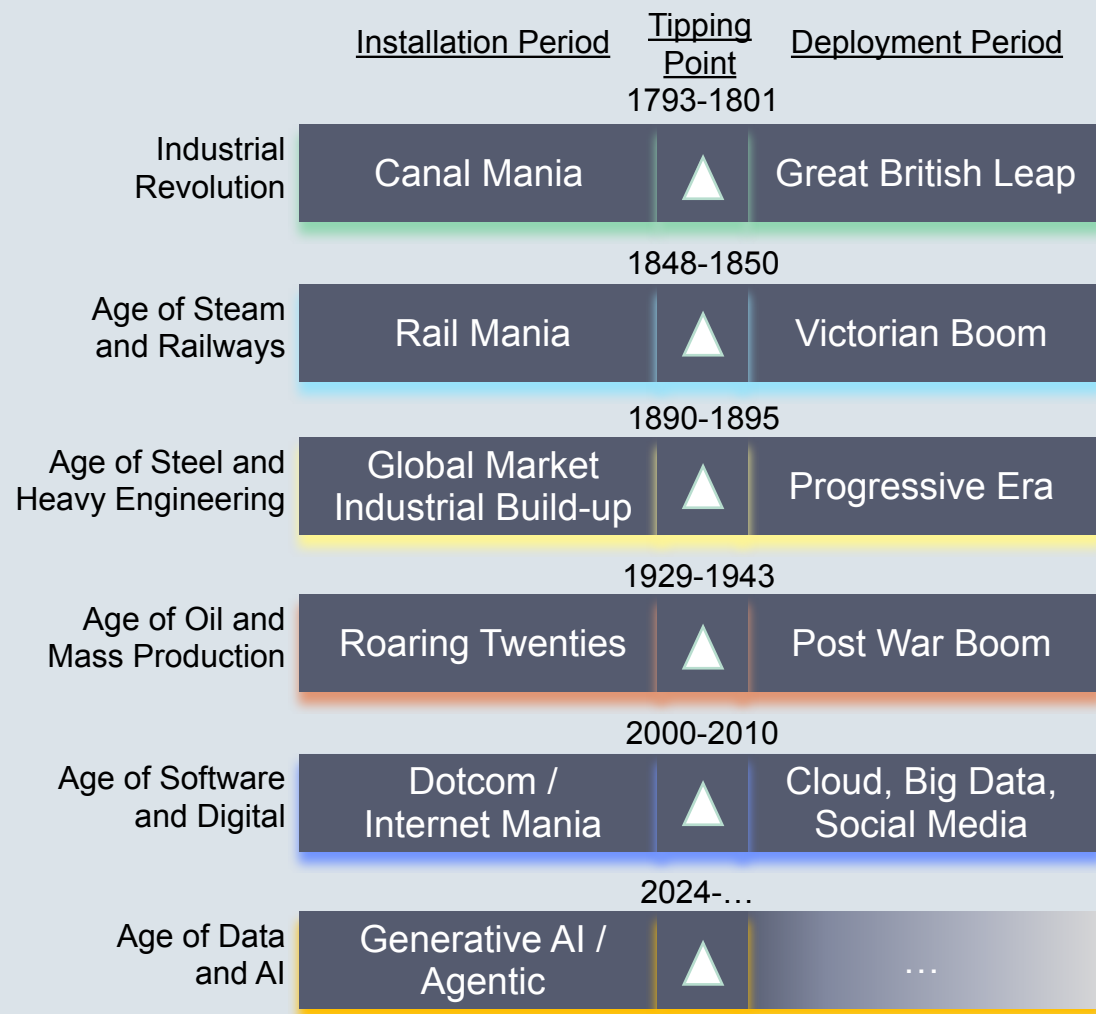
R&D & Infrastructure
\$13.5B

Total Impact
\$116.8B



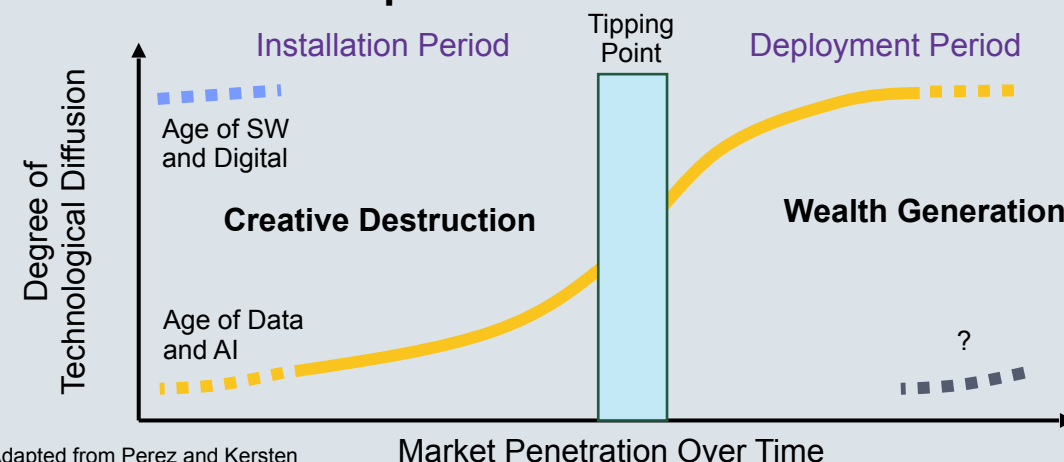
Incrementally Optimizing Cost of Production

Global Economic Transformations



Adapted from Perez and Kersten

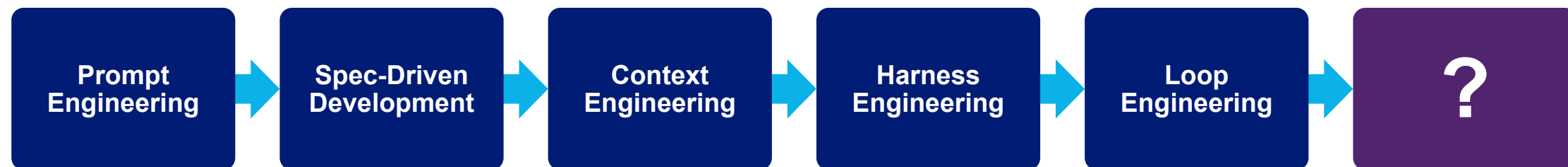
Shape of Transformation



Agentic Engineering → Age of Data and AI	DevOps → Age of SW and Digital
Product Design and Development	Product Delivery (Build, Test, Deploy)
Create new products and services that solve customer problems using hypothesis-driven delivery, modern UX, design thinking	Enable fast flow from development to production and reliable releases by standardizing work, reducing variability, and batch sizes
Feature design and implementation may require work that has never been done before	Integration, test, and deployment must be performed quickly and continuously
Estimates are highly uncertain	Cycle time should be well-known and predictable
Outcomes are highly variable	Outcomes should have low variability
Change Committed into Version Control	

Adapted from Forsgren, Humble, Kim, Willis

Riding the Agentic AI Improvement Waves

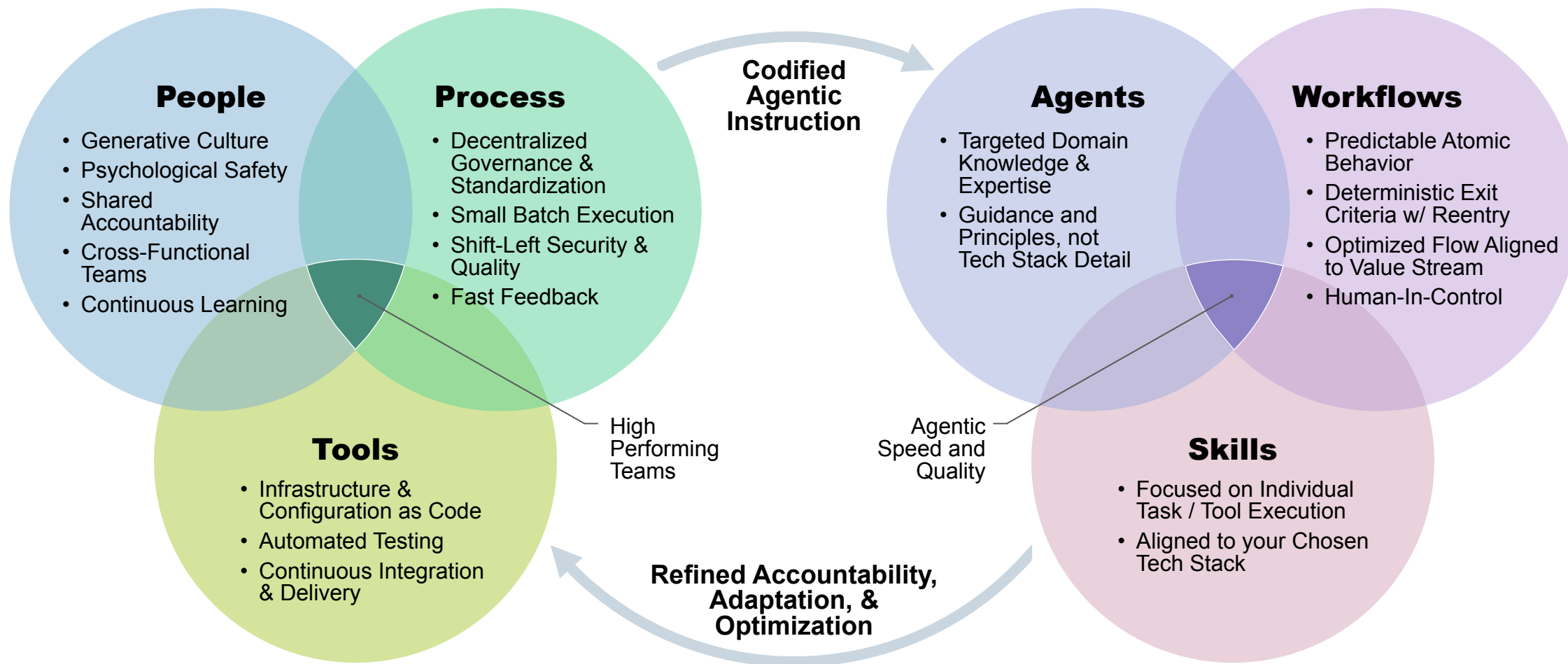


Nondeterminism – The Great Promise / The Great Problem

The Anchorman Challenge:

*They've done studies you know,
60% of the time – it works every time.*

Core Foundations Tailored for Agentic Execution

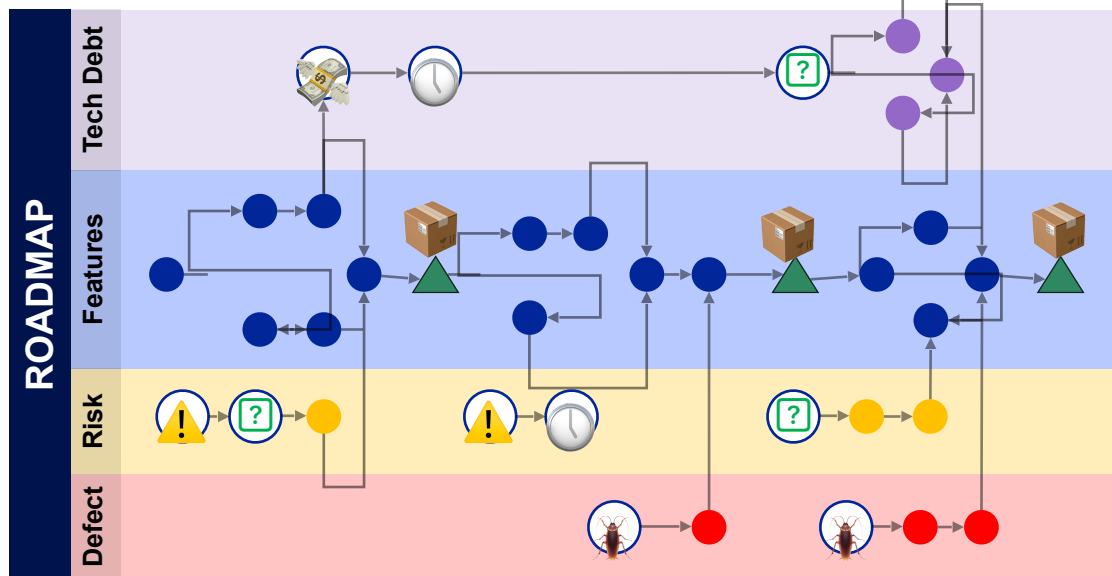


Realized Through: Leadership, Documentation, Training, & Experience

Realized Through: Value Stream Measurement, CM'ed Definitions, Behavioral Validation, & Dependency Management

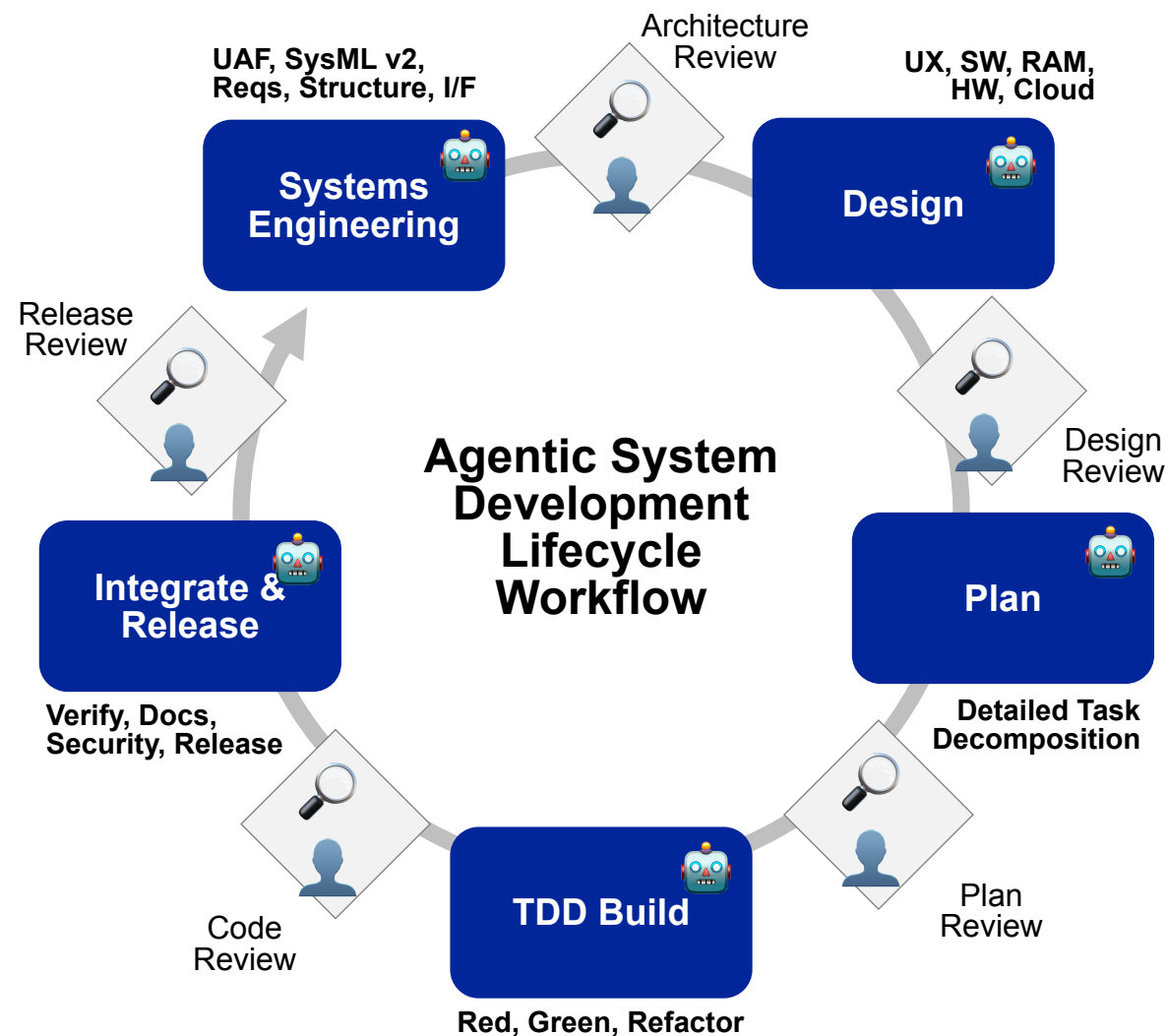
Architecture Driven Engineering

Executable Strategy “Outer Loop”

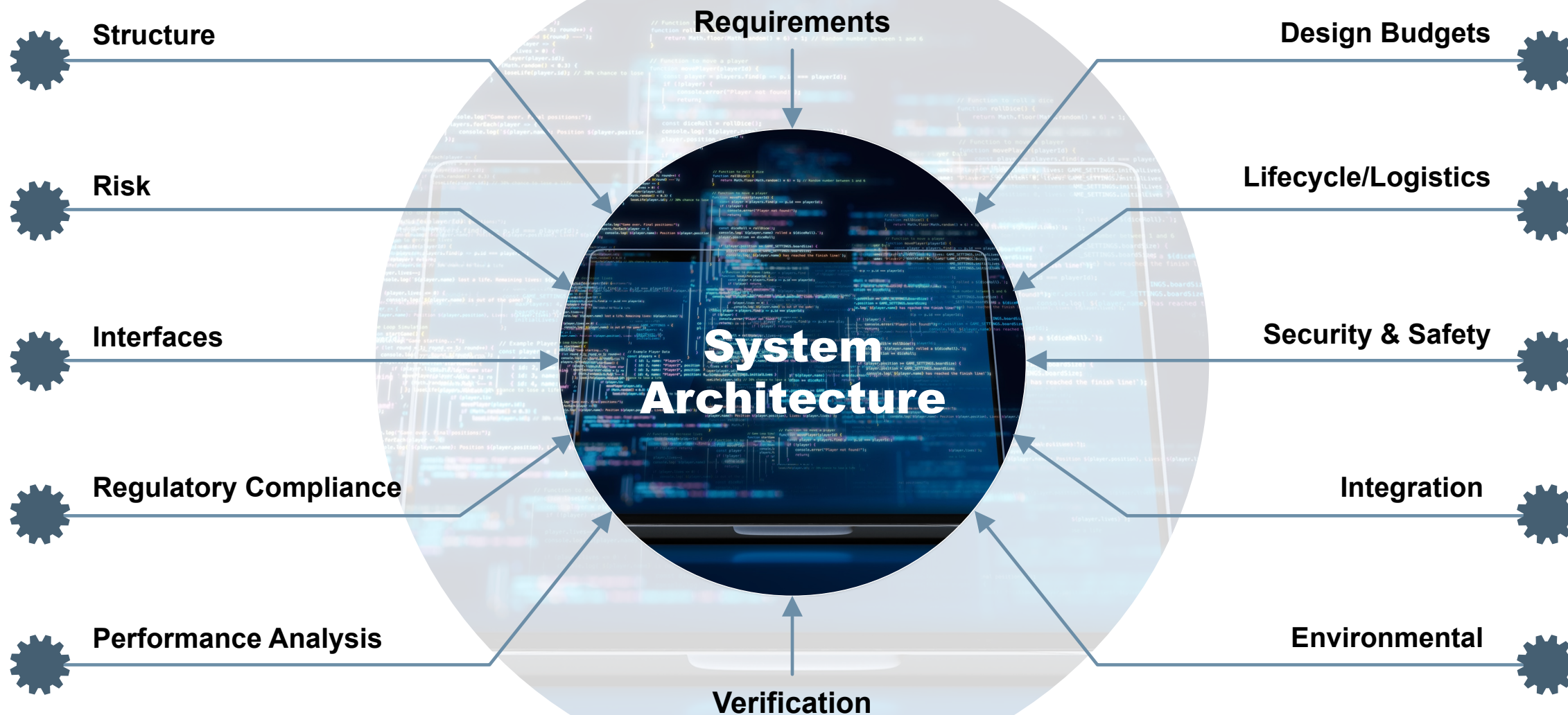


BACKLOG	#	Type	Title	Assignee
	1	●	~~~~~	😊
	2	●	~~~~~	😁
	3	●	~~~~~	👩
	...	●	~~~~~	👤

Iterative Execution “Inner Loop”



Architecture & System Engineering 101



SysML v2 Standard → Architecture-as-Code → Agentic Engineering

Our Agentic Software / IT SDLC Feature Workflow

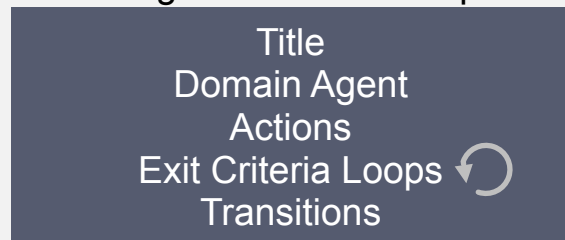
Anatomy of a Workflow

1. Codify the value stream



- Decompose complexity
- Simplify / Linearize
- Eliminate where feasible

2. Define agent execution steps



3. Define human-in-control gates



Agentic Execution

Human-In-Control

Phase 1 Architecture & System Engineering

Needs
Analysis

Logical
Architecture

Physical
Architecture
and Trade Analysis

Architecture
Gate

Phase 2 Multi-Domain and Specialty Design

UX
Design

SW Design

HW / Cloud
Design

RAM
Analysis

Design
Gate

Phase 3 Planning

Current State
Analysis

Target State
Analysis

Task Definition
& Phasing

Plan
Gate

Phase 4 Test Driven Development

Define
Acceptance
Test

Build Minimal
Implementation

Quality
Refactor

Code
Gate

Phase 5 Int / Test, Documentation, & Release

Architecture
Conformance

Cyber
Analysis

Verification
Analysis

Docs

Release
Gate

What We've Observed, Learned, and Measured

Rigor, Discipline, & Flow Drives Rapid / Provable Success ...

- Agentic AI adheres to proven process & practice
 - Small batch execution, throw it away quickly if it feels wrong
 - Dependency version control of your agent, skills, and workflows
- Purposeful design of human / agent team & flow
 - Humans → guidance, oversight, corrective action
 - Workflows → process / Agents → domain knowledge / Skills → task execution
- Human accountability is amplified and auditable
 - Employ formalized human gate reviews with documented approvals

Architecture Driven Human / AI Drives Acceleration ...

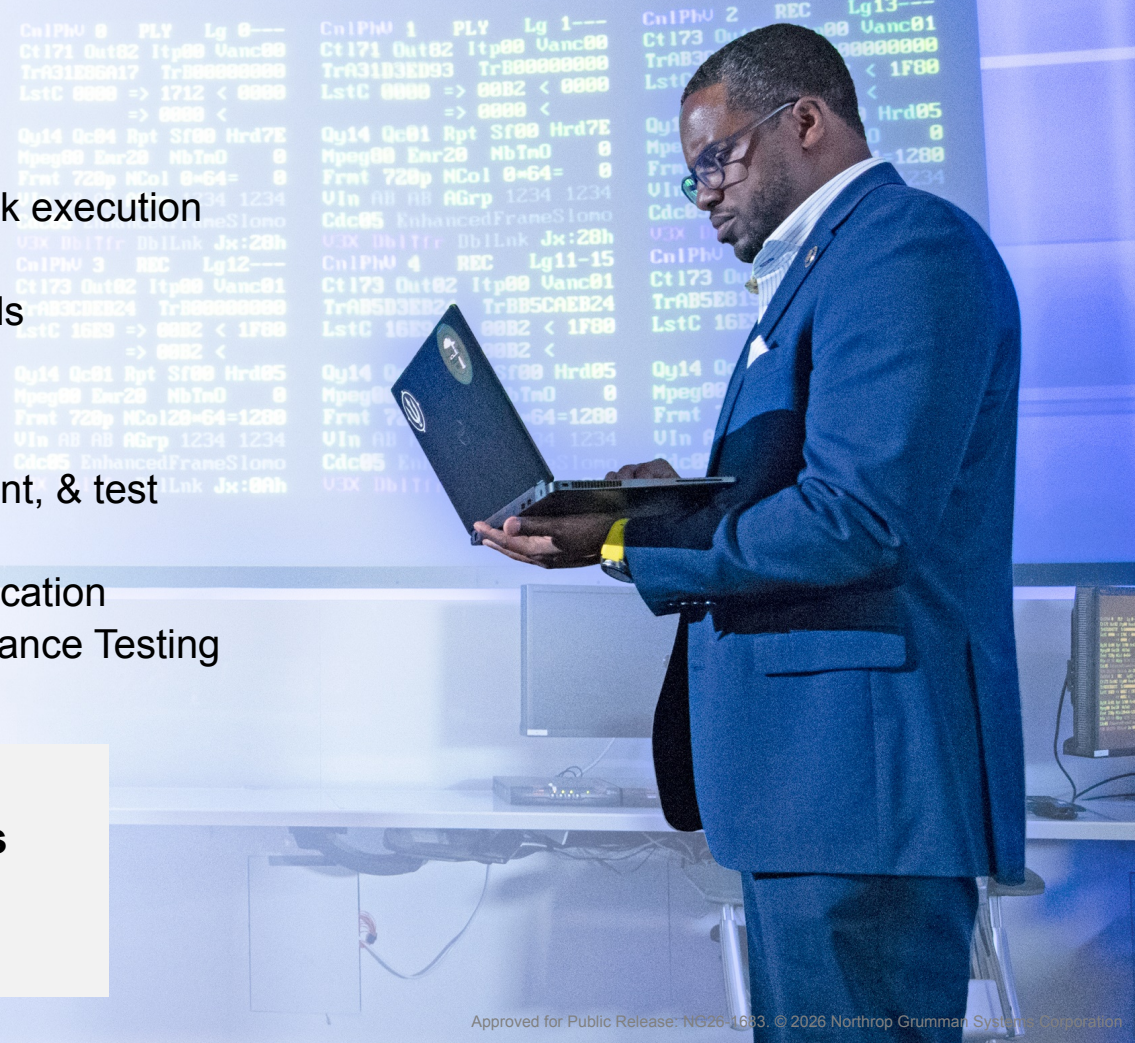
- SysML v2 agent fluency maintains the system architecture baseline
 - Ensure full traceability throughout architecture, design, development, & test
- Quality standards and robust traceability matter
 - Requirements decompose and trace to system structure with verification
 - Requirements (INCOSE + EARS) trace to BDD Automated Acceptance Testing

We can now ...

Build complex new applications / COTS alternatives **in weeks**

Create secure full stack infrastructure deployment **in days**

Find and fix vulnerabilities **in hours**



My Requests to this Community



Embrace Architecture Driven AI Engineering Rigor

- System development & operations truths remain valid
- Agentic control & human accountability

Agentic Version Control & Dependency Management

- We need a standardized AI package manager

Standardize Agentic Workflows

- Develop and mature deterministic agentic workflow definition standards

Continue to Innovate & Share

- US Open-Source
- Community Knowledge Sharing
- Let's talk - tell me what works for you

What do you do when...

Rigor ~~Code~~ is free?

